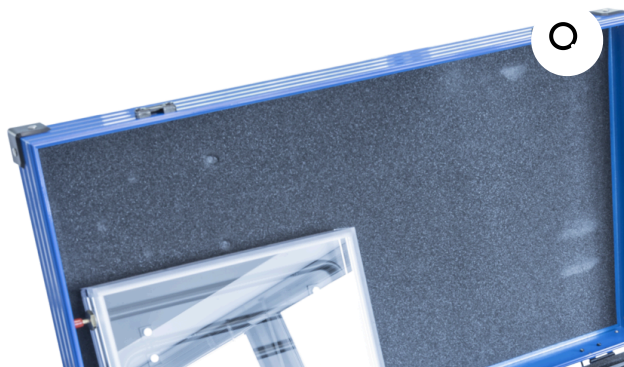




RE16 – Advanced Thermal Energy

The RE16 Advanced Thermal Energy system provides students with a self-contained modular system, covering the different technologies for solar thermal energy transformation on a laboratory scale.

Included in the system are various solar collector systems (including Parabolic reflector and Absorber tube) which can be operated with or without the supplied pump.

PRODUCT CODE: RE16

Categories: RE Series - Renewable Energy, RE16 Thermal Energy

Tags: Concentrated Solar Power), Convective Flow Of Heat , CSP , Electrical Power , Heat Exchange , Heat Radiation , Laboratory Scale , Paraffin Heat Reservoir , Peltier Element , RE16 , Renewable Energy , Solar Thermal Energy Transformation , Thermal Energy , Thermodynamics , Thermoelectric Generator

DESCRIPTION

The RE16 Advanced Thermal Energy system provides students with a self-contained modular system, covering the different technologies for solar thermal energy transformation on a laboratory scale.

Included in the system are various solar collector systems (including Parabolic reflector and Absorber tube) which can be operated with or without the supplied pump.

Additionally supplied are CSP-technology (Concentrated Solar Power) and a Peltier element for the direct transformation into electric energy.

Experimentation includes basics of thermodynamics, heat exchange, absorption of heat radiation and the convective flow of heat.

TECHNICAL SPECIFICATIONS

- 1 x Base unit large
- 1 x Motor module without gear
- 1 x Solar collector
- 1 x Parabolic reflector
- 1 x Absorber tube

- 1 x Lens module
- 1 x Absorber module for lens
- 1 x Absorber B/W
- 1 x Pump module
- 1 x Peltier module
- 1 x Heat exchanger water
- 1 x Heat exchanger paraffin
- 1 x Hose-set
- 1 x Power module
- 2 x Sorting rubber d=65, mark P
- 1 x Propeller
- 2 x Safety test lead, 50cm, red
- 2 x Safety test lead, 50cm, black
- 1 x Lamp housing
- 1 x Bulb infrared 230V
- 1 x Safety plug, black
- 1 x Safety plug, red
- 6 x Safety socket adapter SA 4000 red
- 4 x Safety socket adapter SA 4000 black
- 2 x Digital multimeter
- 1 x Laboratory thermometer
- 1 x Beaker 250ml
- 1 x Temperature measuring sensor
- 1 x Cooling pad
- 1 x Aluminium case

FEATURES & BENEFITS

- Experimentation system for solar thermal energy conversion
- Quantitative experiments for different collector systems
- Flexible and location-independent usage
- Laboratory scale
- Modular design
- Supplied in a self-contained aluminium case
- Includes in-depth manual and predefined experiments

Experimental content

- Absorptivity and reflectivity of different materials
- Focusing of light by a Fresnel lens
- Thermal convection and layering
- Thermal conduction
- Thermal insulation
- Solar thermal collector with pump circulation
- Solar thermal collector with thermosiphon circulation
- Variation of the flow speed



- Collector circuit with heat exchanger
- Collector circuit with paraffin heat reservoir
- Parabolic trough collector with pump cycle
- Defocussing
- Qualitative demonstration of the functional principle
- Investigating the thermoelectric generator
- Quantitative determination of the electrical power

Related curriculums

- Renewable Energies
- Air-conditioning
- Heating
- Energy Efficiency

Other products in the series

- **RE10:** Advanced Photovoltaic Energy
- **RE12:** Advanced Wind Energy
- **RE14:** Advanced Fuel Cell Technology
- **RE18:** Advanced Smart Grid Technology
- **RE24:** Advanced Battery Technology

Operational conditions

- Storage Temperature: -10°C to +70°C
- Operating temperature range: +10°C to +50°C
- Operating relative humidity range: 0 to 95%, non-condensing

Requirements

Electrical supply: 110-230V AC 50-60Hz

- Level and stable work surface

Shipping Specifications

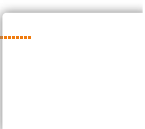
PACKED AND CRATED SHIPPING SPECIFICATIONS

- **Volume:** 0.038m³
- **Gross weigh:** 8.5Kg

Dimensions

TRAY

- **Length:** 0.640m
- **Width:** 0.165m



- **Height:** 0.370m

Order Codes

RE16: Advanced Thermal Energy

YOU MAY ALSO LIKE...



**RE 10 –
ADVANCED
PHOTOVOLTAIC
ENERGY**

[Read more](#)



**RE12 –
ADVANCED
WIND ENERGY**

[Read more](#)



**RE22 –
BIOENERGY**

[Read more](#)



**RE24 –
ADVANCED
BATTERY
TECHNOLOGY**

[Read more](#)



RE20 – BIOFUEL

[Read more](#)



**RE14 –
ADVANCED
FUEL CELL
TECHNOLOGY**

[Read more](#)

Armfield Limited. Unit 10 Headlands Business Park, Salisbury Road, Ringwood, Hampshire BH24 3PB.
England. Tel: +44(0) 1425 478781